ZENER DIODE

Zener Diode Explanation

A Zener Diode, also referred to as a breakdown diode, is a specially doped semiconductor device engineered to function in the reverse direction. When the voltage across a Zener diode's terminals is reversed and reaches the Zener Voltage (also known as the knee voltage), the junction experiences a breakdown, allowing current to flow in the opposite direction. This phenomenon, known as the Zener Effect, is a key characteristic of Zener diodes.

Zener Diode Definition

A Zener diode is a highly doped semiconductor device specifically designed to function in the reverse direction. It is engineered with a wide range of Zener voltages (Vz), and certain types are even adjustable to achieve variable voltage regulation.

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How does a Zener Diode work in reverse bias?

A Zener diode functions similarly to a regular diode when forward-biased. However, in reverse-biased mode, a small leakage current flows through the diode. As the reverse voltage increases and reaches the predetermined breakdown voltage (Vz), current begins to flow through the diode. This current reaches a maximum level determined by the series resistor, after which it stabilizes and remains constant across a wide range of applied voltages.

There are two types of breakdowns in a Zener Diode: **Avalanche Breakdown and Zener Breakdown**.

Avalanche Breakdown in Zener Diode

Avalanche breakdown occurs in both normal diodes and Zener diodes when subjected to high reverse voltage. When a significant reverse voltage is applied to the PN junction, the free electrons gain enough energy to accelerate at high velocities. These high-velocity electrons collide with other atoms, causing the ejection of additional electrons. This continuous collision process generates a large number of free electrons, resulting in a rapid increase in electric current through the diode. In the case of a normal diode, this sudden surge in current could permanently damage it. However, a Zener diode is specifically designed to withstand avalanche breakdown and can handle the sudden current spike. Avalanche breakdown typically occurs in Zener diodes with a Zener voltage (Vz) greater than 6V.

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Zener Breakdown in Zener Diode

When the reverse bias voltage applied to a Zener diode approaches its Zener voltage, the electric field within the depletion region becomes strong enough to attract and remove electrons from their valence band. These valence electrons, energized by the intense electric field, break free from their parent atoms. This phenomenon takes place in the Zener breakdown region, where even a slight increase in voltage leads to a rapid surge in electric current.

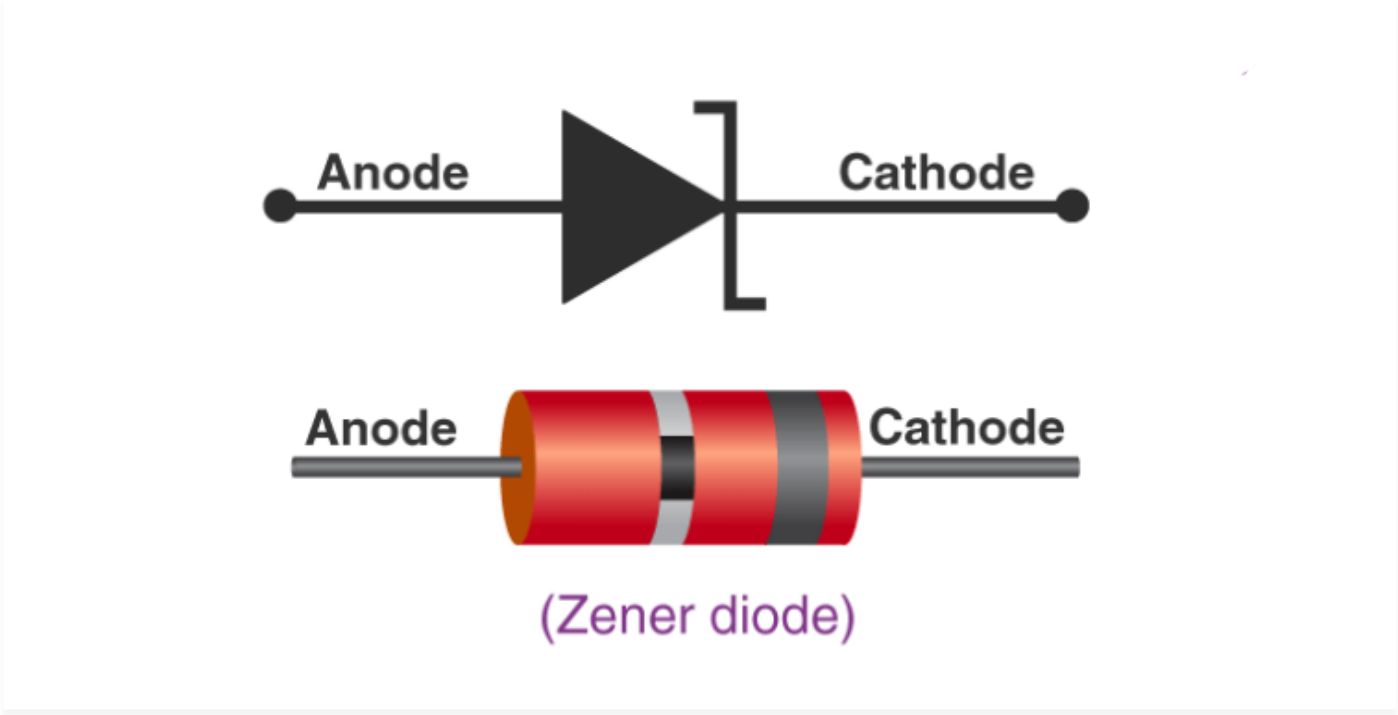
Avalanche Breakdown vs. Zener Breakdown

The Zener effect is predominant in voltages up to 5.6 volts, while the avalanche effect becomes more prominent beyond that threshold. Although both effects are similar, the distinction lies in the fact that the Zener effect is a quantum phenomenon, whereas the avalanche effect involves the movement of electrons in the valence band, similar to an electric current. The avalanche effect allows a larger current through the diode compared to what a Zener breakdown would permit.

ZENER DIODE

Circuit Symbol of Zener Diode

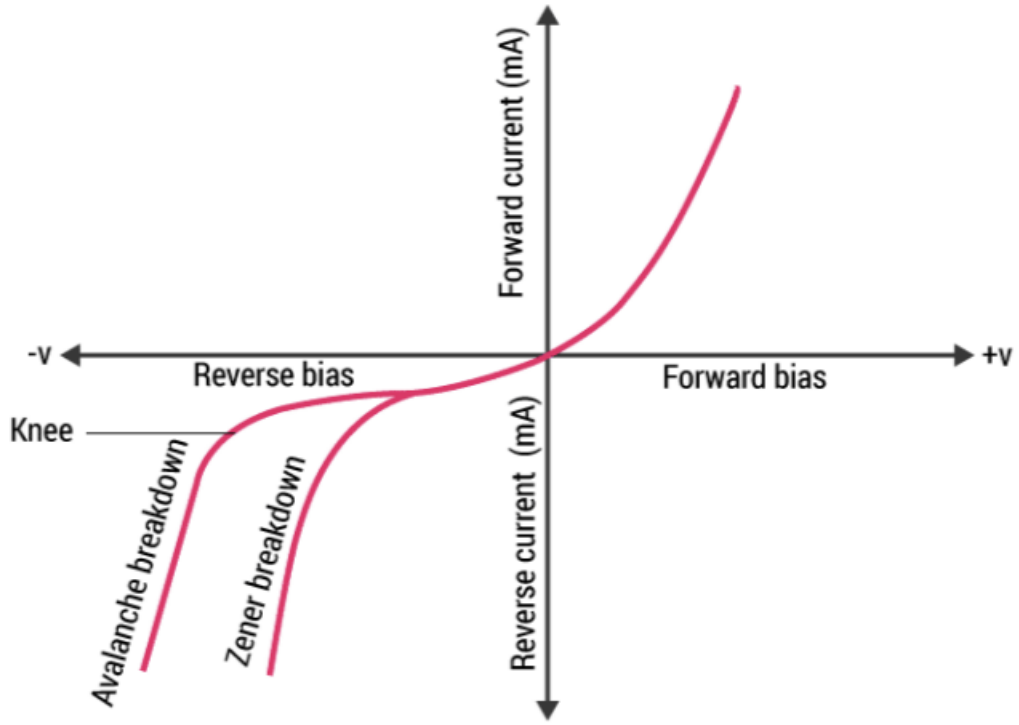
Zener diodes come in various packaging options, depending on their power dissipation requirements. Some are designed for high-power applications, while others are available in surface mount formats. The most commonly used Zener diode is packaged in a small glass enclosure, with a distinctive band indicating the cathode side of the diode.



ZENER

V-I Characteristics of Zener Diode

The diagram given below shows the V-I characteristics of the Zener diode.



When reverse-biased voltage is applied to a Zener diode, it allows only a small amount of leakage current until the voltage is less than Zener voltage.

The V-I characteristics of a Zener diode can be divided into two parts as follows:

(i) Forward Characteristics

(ii) Reverse Characteristics

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(i) Forward Characteristics
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Forward Characteristics of Zener Diode

The first quadrant in the graph represents the forward characteristics of a Zener diode. From the graph, we understand that it is almost identical to the forward characteristics of P-N junction diode.

Reverse Characteristics of Zener Diode

When a reverse voltage is applied to a Zener voltage, a small reverse saturation current Io flows across the diode. This current is due to thermally generated minority carriers. As the reverse voltage increases, at a certain value of reverse voltage, the reverse current increases drastically and sharply. This is an indication that the breakdown has occurred. We call this voltage breakdown voltage or Zener voltage, and Vz denotes it.

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Zener Diode Specifications

Some commonly used specifications for Zener diodes are as follows:

- **Zener/Breakdown Voltage** – The Zener or the reverse breakdown voltage ranges from 2.4 V to 200 V, sometimes it can go up to 1 kV while the maximum for the surface-mounted device is 47 V.
- **Current Iz (max)** – It is the maximum current at the rated Zener Voltage (Vz – 200μA to 200 A)
- **Current Iz (min)** – It is the minimum value of current required for the diode to break down.
- **Power Rating** – It denotes the maximum power the Zener diode can dissipate. It is given by the product of the voltage of the diode and the current flowing through it.
- **Temperature Stability** – Diodes around 5 V have the best stability
- **Voltage Tolerance** – It is typically ±5%
- **Zener Resistance (Rz)** – It is the resistance to the Zener diode exhibits.

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Application of Zener Diode

Following are the applications of Zener diode:

Zener diode as a voltage regulator:

The zener diode is used as a Shunt voltage regulator for regulating voltage across small loads. The Zener diode is connected parallel to the load to make it reverse bias, and once the Zener diode exceeds knee voltage, the voltage across the load will become constant. The breakdown voltage of Zener diodes will be constant for a wide range of currents.

Zener diode in over-voltage protection:

When the input voltage is higher than the Zener breakage voltage, the voltage across the resistor drops resulting in a short circuit, this can be avoided by using the Zener diode.

ZENER DIODE

Frequently Asked Questions – FAQs

Q1 **How do you define Zener diode?**

Zener diode is a semiconductor device that enables the flow of current in both forward and reverse directions.

Q2 **Why is Zener Diode used a regulator?**

Zener diode is utilized as a shunt voltage regulator. It is connected in parallel with the load, operating in reverse bias. Once the Zener diode exceeds its knee voltage, it maintains a constant voltage across the load.

Q3 **Does Zener Diode exhibit a controlled breakdown?**

Yes, Zener Diode does exhibit a controlled breakdown.

Q4 **Does Zener Diode work in forward-biased mode?**

No, a Zener diode does not function in forward-biased mode. It is designed to operate in reverse bias for its intended applications

Q5 **What is the difference between a Zener diode and a normal diode?**

The primary difference between a Zener diode and a normal diode lies in their current conduction. A normal diode allows current to flow in only one direction, while a Zener diode permits current to flow in both directions.

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Q6 **What is a Zener diode also called as?**

A Zener diode is also called as a breakdown diode.

Q7 **Who was the first person to describe the electrical properties of the Zener diode?**

Clarence Melvin Zener was the first person to describe the electrical properties of the Zener diode.

Q8 **What are the types of breakdowns for a Zener Diode?**

There are two types of breakdowns for a Zener Diode:

- Avalanche Breakdown
- Zener Breakdown

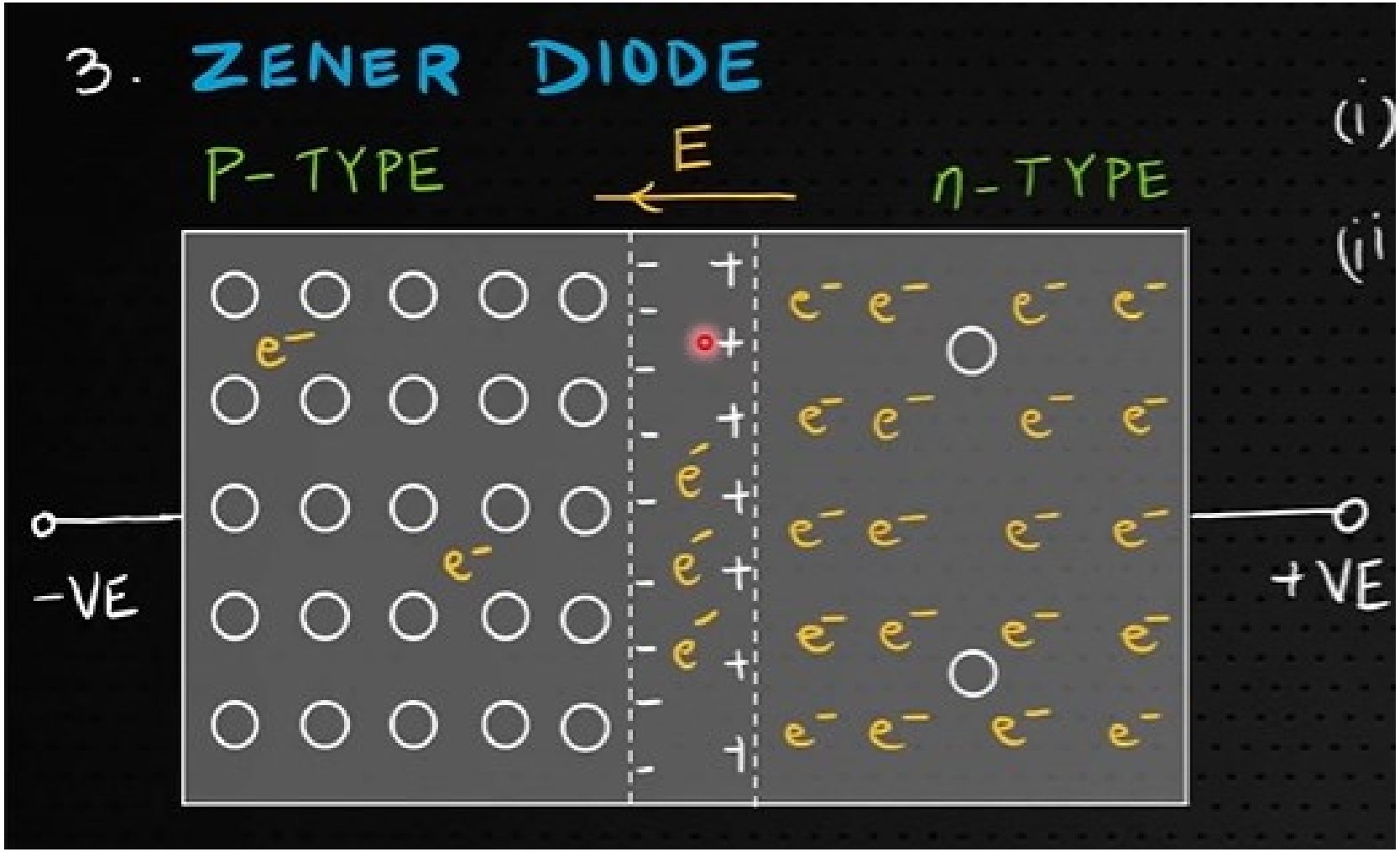
Q9 **What is the voltage tolerance of a Zener diode?**

Voltage tolerance of a Zener diode is typically ±5%.

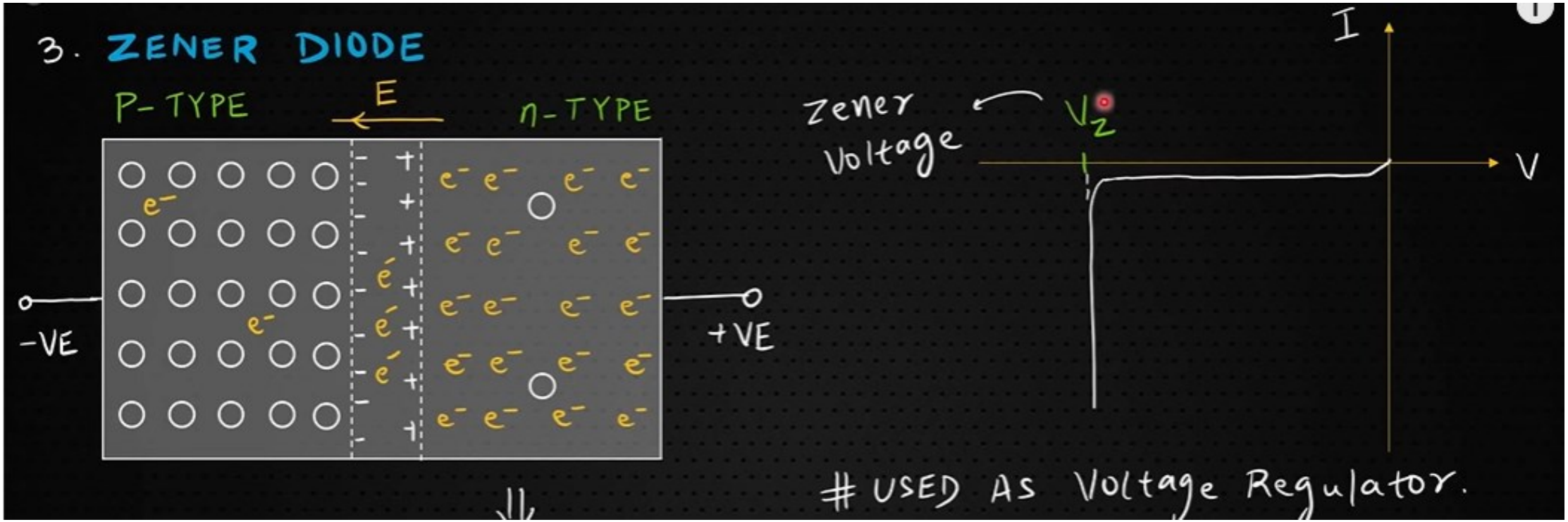
Q10 **State true or false: Zener diode is used as a shunt voltage regulator.**

Yes, for regulating voltage across small loads, the Zener diode is used as a shunt voltage regulator.

ZENER DIODE



ZENER DIODE


3. ZENER DIODE
P-TYPE
E
n-TYPE
-VE
+VE
I
V
Zener Voltage
V_Z
USED AS Voltage Regulator.